

Project Documentation

Final Report

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Project Name	Automated Network Request Management in ServiceNow

1. INTRODUCTION

1.1 Project Overview

The Automated Network Request Management System is a ServiceNow cloud-native workflow automation solution designed to streamline the lifecycle of enterprise network service intake, verification, approval, and task execution. Traditional network management workflows rely heavily on manual coordination via unstructured email threads, verbal requests, and paper forms across network engineers, managers, and requesters. This leads to operational backlogs, missing configuration parameters, and compliance risks.

This project modernizes the end-to-end process using the Service Portal (/sp), Flow Designer, custom table Database Tables (u_database_table), dynamic Catalog UI Policies, automated approval gating, task generation, role-based access control (ACLs), and email notifications. The system guarantees that every network request is captured with complete data integrity, routed through structured manager approvals, and tracked transparently.

1.2 Purpose

The primary objectives of this project include:

- Automate network connection requests (New and Existing connections) via a self-service catalog item.
- Eliminate incomplete submissions using dynamic Catalog UI Policies that reveal fields like 'Existing Id' conditionally.
- Persist all request, pricing, and location data into a dedicated custom table Database Tables (u_database_table) with auto-numbering (prefix DBT).
- Automate managerial and security approvals using zero-code Flow Designer logic.

- Trigger immediate outbound email notifications at each milestone (Submission Receipt, Approval Notice, Rejection Notice).
- Provide operational audit trails and ensure role-based data security through ServiceNow ACLs.

2. IDEATION PHASE

2.1 Problem Statement

Traditional enterprise network provisioning suffers from three primary bottlenecks: (1) Opaque request progression, leaving requesters anxious and driving repeated manual follow-ups; (2) High engineering triage overhead, requiring network engineers to chase missing technical details; and (3) Non-compliance risks from unrecorded, ad-hoc network configurations without formal manager sign-off.

2.2 Empathy Map Canvas

The empathy map analysis evaluates the requester's perspective:

- Says: I need to establish network access for my workstation and confirm whether it is a New or Existing Connection.
- Thinks: Will this request sit in a backlog for days? I hope my contact details auto-fill correctly.
- Does: Navigates to /sp, fills out Requested For, Phone, Email, Connection Type, Payment Mode (UPI/Card), Address, and submits.
- Feels: Anxious about project delivery delays before automation; relieved and confident upon receiving an instant tracking receipt via email.
- Pains: Lack of approval visibility, repetitive data entry, and manual triage delays.
- Gains: Rapid turnaround, self-service clarity, and automated lifecycle notifications.

2.3 Brainstorming & Idea Prioritization

Alternative architectures were assessed during ideation: (1) Using the default Incident table was rejected because network requests represent planned service requisitions rather than unplanned outages; (2) A custom-coded scoped application was rejected due to heavy maintenance overhead. A low-code Flow Designer architecture integrated with the custom table Database Tables (u_database_table) was selected for its high maintainability, rapid delivery, and seamless integration with sysapproval_approver.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey is structured across five sequential stages: Entice (discovering 'Network Request' on the Service Portal), Enter (filling dynamic inputs including Connection Type, Payment Mode, and Address), Engage (asynchronous Flow execution, record insertion, and manager review), Exit (network engineer configuration and task completion), and Extend (automated completion email and connectivity verification).

3.2 Solution Requirements

Functional Requirements (FR) cover six core domains: Portal Intake (FR-1), Dynamic Form Validation (FR- 2), Database Tables Persistence (FR-3), Flow Designer Automation (FR-4), Approval Governance (FR-5), and Outbound Email Notifications (FR-6).

Non-Functional Requirements (NFR) enforce platform quality attributes: Usability (NFR-1), TLS 1.3 / AES- 256 Security & ACLs (NFR-2), Data Integrity (NFR-3), sub-2.0s Performance (NFR-4), 99.98% High Availability (NFR-5), and Scalability (NFR-6).

3.3 Data Flow Diagram (DFD Level 1)

The DFD Level 1 illustrates the pipeline between user interactions, the workflow engine, approval records, and custom table persistence.

3.4 Technology Stack

The solution is built on ServiceNow Enterprise Cloud Infrastructure (Personal Developer Instance - PDI), utilizing AngularJS and Bootstrap for Service Portal rendering, client-side Catalog UI Policies, Flow Designer for business logic orchestration, MariaDB relational database hosting Database Tables (u_database_table), and the platform Outbound SMTP Mailer.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The solution directly resolves traditional operating bottlenecks: unstructured email requests are replaced by a guided portal form; missing technical inputs are eliminated through mandatory dynamic UI policies; and unrecorded manual approvals are replaced by automated Flow Designer governance linked to sysapproval_approver.

4.2 Proposed Solution

The solution consists of four primary components:

- Service Catalog Form: Captures Requested For, Opened on behalf of, User Name, Phone, Email, Mobile, Type of Connection (New/Existing), Total Amount, Mode of Payment (UPI/Card), Address, and attachments.
- Declarative UI Policies: Dynamically display the 'Existing ID' field only when 'Existing' is selected.
- Dedicated Table (Database Tables - u_database_table): Stores submitted parameters with auto-numbering (prefix DBT).
- 10-Step Flow Designer Workflow: Triggered on catalog submission to extract variables, insert records, dispatch receipt emails (Step 3), trigger manager approvals (Step 4), and execute approved (Steps 6 & 7) or rejected (Steps 9 & 10) branch updates.

4.3 Solution Architecture

The architecture is structured across modular enterprise tiers:

- Service Portal (/sp) & Service Catalog (Intake & Presentation)
- Catalog UI Policies & Client Scripts (Dynamic Validation)
- Database Tables (u_database_table) (Custom Relational Persistence)
- Flow Designer 10-Step Workflow (Business Logic Tier)
- sysapproval_approver & Role-Based ACLs (Governance & Security Tier)
- Outbound SMTP Mailer (Communications Tier)
- sys_attachment (Binary File Repository)

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Development was executed using Agile Scrum methodology across three targeted sprints over an 18-day schedule, completing 30 total story points with an average velocity of 10.0 story points per sprint (1.67 story points per day):

Sprint 1: Intake & Schema Setup (10 Points) - Built Network Request catalog item, configured contact variables, UI policies (Existing ID), and created Database Tables (u_database_table).

Sprint 2: Workflow Automation & Approvals (11 Points) - Configured Flow Designer trigger, variable extraction, record insertion, receipt email (Step 3), and manager approval action (Step 4).

Sprint 3: Branching, Communications & UAT (9 Points) - Configured Step 5 If logic, Approved branch (Steps 6 & 7), Rejected branch (Steps 9 & 10), table ACLs, and final UAT sign-off.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

The following functional capabilities were verified with a 100% pass rate across 47 test cases:

- Catalog item presentation and search indexing on Service Portal (/sp).
- Catalog variable validation (mandatory checks on Address, Amount, Connection Type).
- Dynamic UI policy responsiveness (showing Existing ID on Existing connection).
- Flow Designer trigger execution upon Service Catalog submission.
- Record creation in Database Tables (u_database_table) with auto-numbering.
- Manager approval routing and link to sysapproval_approver.
- Conditional branching: State updates to 'Approved' or 'Rejected'.
- Automated outbound email dispatches (Receipt, Approval Notice, Rejection Notice).
- Table-level ACL enforcement securing requester data.

6.2 Performance Testing

Performance benchmarking evaluated operational latency against enterprise SLAs across nine scenarios (PTC-001 to PTC-009):

- Portal Catalog Page Load: 1.18s (SLA < 2.0s)
- Client UI Policy Dynamic Field Evaluation: 32 ms (SLA < 100 ms)
- Database Tables (u_database_table) Record Insertion: 380 ms (SLA < 1.0s)
- Outbound Receipt Email Generation: 1.45s (SLA < 3.0s)
- Approval Generation & Linking: 620 ms (SLA < 1.5s)
- End-to-End Flow Designer Execution: 3.42s (SLA < 5.0s)

- Overall Performance Safety Margin: 50.6% below maximum allowable latency ceilings.

7. RESULTS

7.1 Output Screenshots & Execution Walkthrough

The developed solution successfully automated the complete network service request lifecycle. Keyoperational walkthrough steps include:

1. Service Portal Intake: Requesters browse /sp, select Network Request, and experience real-time dynamic UI policy behavior where Existing ID appears only for existing lines.
2. Database Persistence: Submission inserts a record into Database Tables (u_database_table) with unique auto-numbering (e.g., DBT0001001) and initial state 'Awaiting Approval'.
3. Flow Designer Automation: Workflow Studio execution details confirm instantaneous variable extraction (Step 1), record generation (Step 2), receipt email dispatch (Step 3), and approval generation (Step 4).
4. Governance & Branching: Approvers review the request; on approval, Step 6 updates the table status to 'Approved' and Step 7 dispatches an approval notification to the requester.
5. Secure Access: Configured ACLs restrict regular users to their own submitted records while granting fulfillers administrative oversight.

8. ADVANTAGES & DISADVANTAGES

Advantages:

- 100% Low-Code Architecture: Built entirely within Flow Designer and declarative UI Policies, eliminating custom script debt.
- Reduced Turnaround Time: Automated intake and approvals cut cycle times from days to hours.
- Eliminated Missing Information: Mandatory client-side validations ensure complete technical and address data upfront.
- Centralized Governance & Auditing: sysapproval_approver linkage creates an immutable audit trail for IT Compliance.

- **Enhanced Transparency:** Automated emails keep requesters informed at every stage without chasing helpdesk agents.

Disadvantages & Constraints:

- **ServiceNow Platform Dependency:** Requires an active instance; PDIs hibernate during inactivity.
- **Downstream Manual Fulfillment:** Physical port wiring and switch patching rely on manual tasks by network engineers.
- **Initial Administrator Learning Curve:** Configuring custom tables, Flow Designer actions, and ACLs requires platform expertise.

9. CONCLUSION

The Automated Network Request Management system successfully demonstrates how the ServiceNow platform can modernize and standardize network service delivery. By replacing error-prone manual email threads with dynamic catalog forms, automated workflow routing, and structured database persistence, the solution eliminates operational bottlenecks, improves user satisfaction, and ensures strict adherence to network security policies. The project achieved a 100% UAT pass rate and confirmed robust performance readiness for enterprise deployment.

10. FUTURE SCOPE

Future enhancements include:

- **Zero-Touch Provisioning:** Integrate Flow Designer with Cisco DNA Center, Ansible, or Infoblox via Integration Hub spokes for automated VLAN and port assignment.
- **Virtual Agent Chatbot Intake:** Enable conversational request submission via Microsoft Teams or Slack using natural language processing.
- **Active Directory & CMDB Integration:** Automatically bind network requests to CI configuration items (routers, switches) in the ServiceNow CMDB.
- **Executive Performance Dashboards:** Implement Performance Analytics to track fulfillment trends, average cycle times, and SLA compliance.

11. APPENDIX

- Implementation Artifacts: Catalog Item 'Network Request', Flow Designer workflow 'Network Request', Custom Table 'u_database_table' schema definitions, and Catalog UI Policies.
- Platform Instance URL: <https://dev425212.service-now.com/sp>
- Project Demo Video & Documentation Repository:
<https://drive.google.com/file/d/1mFeGC3SjjQM5BDIedcjwHEFXJ9Uh-svD/view?usp=sharing>